Climate Catalyst



GREEN PUBLIC PROCUREMENT CASE STUDY: The Buy Clean California Act

About this case study

Twenty seven per cent of India's steel demand comes from public construction and infrastructure projects. With steel production expected to triple by 2050, this represents an opportunity to drive sectoral transformation while supporting economic growth.

Green Public Procurement (GPP) isn't just another policy tool, it's how governments worldwide are creating markets for low-carbon steel. In this series, we're exploring how Green Public Procurement is being used and considered by governments across the world to help drive the creation of a market for green steel products. It aims to offer examples of existing and incoming policy which could be relevant in the Indian context.

This is part of a series of case studies which we will release over the course of 2025. You can access all of the case studies at https://bit.ly/GPPinIndia

About Climate Catalyst

Climate Catalyst works to compel our political leaders to take action at speed and scale. We do this by galvanising the collective power of businesses, investors and civil society to influence and accelerate policy change. Working behind the scenes we bring together new and diverse stakeholders to identify opportunities with the greatest potential for collaboration, share ideas and learnings and deliver creative campaigns that build power and secure decisive action by governments that reduce greenhouse gas emissions. Our vision is for a just, prosperous world in which global temperature rise is limited to 1.5°C.

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Introduction

India is poised as one of the only bright spots among flattening global steel demand. The country will see a rapid expansion of capacity in the years ahead, when most of the world replaces or retrofits existing capacity. Given this unique opportunity and its climate targets, India has taken its first steps to embed climate and emissions considerations in the steel sector.

The government has released a roadmap for a low-carbon steel sector, a green steel taxonomy and has even made a small budget allocation for innovations in lower-carbon steel making in the country. The Indian government's newly announced green steel taxonomy leverages the rapid infrastructure ramp-up in housing, railways and highways, where a significant share of steel purchased will be via public procurement. In this context, the public procurement of low-carbon steel in California via its Buy Clean program could provide valuable learnings for Indian policymakers.

The Buy Clean California Act (BCCA) was the first legislation of its kind in the United States that integrated climate considerations in public procurement processes. Passed in 2017, it targets carbon emissions from four key construction materials – structural steel, concrete reinforcing steel, flat glass and insulation. It requires these materials used in public works projects to meet an emission standard.

The BCCA helped lay the groundwork for Buy Clean policies in other US states, and informed Buy Clean regulations and emerging policy at the Federal level. Its technical approach – using Environmental Product Declarations (EPDs) – has lessons for Indian policymakers on GPP, as does the cooperation between unions, industry and regulators in designing a policy that is both ambitious and pragmatic.



About the Buy Clean California Act

When did it come into force? The law, AB 262, was passed in 2017, followed by a period of data collection. Emissions standards went into effect in July 2022.

Who does it apply to? Contractors bidding on state construction projects must provide information on the greenhouse gas emissions associated with those four materials.

How does it work? It requires materials manufacturers to conduct a product lifecycle assessment to calculate the global warming potential (GWP), expressed as carbon dioxide equivalent per metric tonne of material, of their products. They must produce a verified Environmental Product Declaration (EPD), either themselves or through an independent consultant, to document that GWP. The EPD must be independently verified.

Only those materials with a GWP below thresholds set by California's Department of General Services (DGS), typically set at the industry average, can be used in California state-commissioned public works projects (see Table).

Every three years from 1 January 2025 the DGS reviews the GWP thresholds for each material and may revise them downwards to reflect industry improvements.

TABLE: CURRENT EPD STANDARDS FOR STEEL PRODUCTS

Material	Unfabricated GWP
Hot-rolled sections	1,010kg $C0_2e$ per metric tonne
Hollow structural sections	1,710kg $C0_2e$ per metric tonne
Steel plate	1,490kg $C0_2e$ per metric tonne
Concrete reinforcing steel	890kg $C0_2e$ per metric tonne

Source: DGS (2025), Buy Clean California Act global warming potential adjustment study

The genesis of the policy

The concept of Buy Clean legislation was first proposed by a coalition of trade unionists and environmentalists, working through a group called the Blue Green Alliance. At the same time, a steel mill in California which had invested heavily in pollution control equipment was looking for commercial advantage over imported steel. The management approached the steel workers union to explore how they could benefit from its steel's reduced climate impact.

The Act's advocates found a champion for their legislation in Assembly Member Rob Bonta, who helped build bipartisan support for the proposal. He was able to overcome opposition from some of the agencies, including DGS, that would be required to implement the Act, who were concerned about the novelty of the approach, and from initial reservations from Governor Jerry Brown, who initially shared those concerns. Governor Brown signed the BCCA into law on 15 October 2017.

Putting policy into practice

The BCCA required California's DGS to set thresholds for the global warming potential of each material covered by the legislation. However, that process took longer than anticipated. The DGS had "reasonable concerns" about the challenges in implementing the legislation, says Mike Williams, a senior fellow at the Center for American Progress, meaning that it had to extend the deadline for introduction of requirements to submit EPDs from July 2021 to July 2022.

This delay notwithstanding, the BCCA requirements successfully came into force, with the design process broadly supported by industry trade associations. In steel, the American Institute of Steel Construction (AISC) recognised the potential of the Act to improve the sector's competitiveness, and worked with DGS to understand the nature of the industry and what thresholds would be appropriate. "It is useful to have the trade association's expertise during policy implementation if they see benefit for their members and are acting on a good-faith basis," says Meghan Lewis, program director at the Carbon Leadership Forum (CLF), in Seattle, Washington State. "AISC saw they could support their members by helping DGS."

One key industry contribution was to encourage the introduction of three sub-categories for steel – hot-rolled sections, hollow structural section and plate – because a single GWP limit for steel would not have been appropriate for different steel types, given the varying emissions levels in the different manufacturing processes involved. This is a key consideration for other jurisdictions considering the introduction of similar legislation. In the Indian context, the broad criteria set by its green steel taxonomy provides a good starting point. But, over time, a refining of those standards, depending on the sector, is going to be necessary, perhaps by acknowledging the share of scrap in low-carbon steel.



Taking the EPD route

EPDs are akin to a certificate that ensures information transparency and are comparable internationally. As commonly used tools for a variety of products, often combined with ecolabelling in many markets,[1] EPDs provide facility-specific performance data, verified by third parties. Compared to other systems in Europe at the time like eco labels or the ISO 14001 certification which use point-scoring to evaluate the environmental impact of inputs, EPDs offered a simpler approach by stating a product's GWP directly. This allowed buyers to easily assess whether a product met the threshold, providing a clear 'yes/no' standard that reduced administrative burden and minimised disputes between producers and purchasers.

By introducing a GWP threshold for various construction materials, communicated via EPDs, the BCCA provides clear guidance for industry, argues Nancy Gillis, a Zurich-based industrial decarbonisation expert. "Industry loves clarity," she says. The BCCA creates a threshold for GWP below which suppliers won't be eligible for contracts. "That's the biggest signal – that's when industry takes note," she says.

When legislators were drafting the bill, relatively few EPDs existed in the US, notes Lewis, but there were some leading producers who had created them to demonstrate the environmental credentials of their products. "There were at least a handful of suppliers who demonstrated the case for EPDs as a way to calculate emissions and prove they were lower carbon than their competitors," she says. Similarly in India, some companies are already familiar with, and issuing their own EPDs. Tata Steel, for example, issued its first EPD for rebar steel in 2022, using it to help it compete for projects in the construction sector globally. In 2022, it also acquired an EPD for structural steel hollow sections. It would be easy to see how this uptake could be widened under any new GPP policy in India.

EPDs do, however, impose costs on participating businesses. A steel plant in the US will need to spend from US$10,000 to US$25,000 on an EPD, a process which needs to be repeated every five years. While the BCCA did not provide subsidies for EPD drafting, the federal Environmental Protection Agency under the Biden Administration introduced a US$200 million grant programme to support EPD drafting nationally; however, this programme was cancelled by the successor Trump Administration. Providing subsidies is likely to be a key consideration in the Indian context, given heterogeneity among domestic steel producers, not all of whom will be able to bear the compliance burden.

Building flexibility into threshold-setting

Any green procurement policy needs to ratchet down to deliver environmental impact, as well as respond to improving industry performance and technological innovation. BCCA builds in a three-year review process, the first of which was carried out in January 2025. It made one change to steel GWPs: reducing the maximum GWP for unfabricated concrete reinforcing steel to 755 kg CO^2e, and to 778kg for fabricated product, from 890 kg and 920 kg respectively.

There is a balance to be struck, says Anish Tilak, at the Rocky Mountain Insitute's (RMI's) Carbon-free Buildings Program, between writing detailed legislation and ensuring flexibility in implementation: "If you include too many details for the policy in the legislation, it makes it more challenging to change after the fact." The ideal, he adds, is to create a framework for the policy with the legislation and empower an agency to develop the policy's coverage, targets, etc. over time. In India, the National Institute of Secondary Steel Technology (NISST) has been designated as the nodal agency for the green steel taxonomy to carry out monitoring, reporting, verification (MRV) and certification of low-carbon steel, and could potentially continue in that role. The priority now is to establish a consistent policy framework with progressively increasing ambition against the benchmark set by the green steel taxonomy unveiled by the Ministry of Steel in December 2024

1 An ecolabel is a designation that a product or service meets specific environmental performance criteria, without providing the same level of detail as to its precise performance.



BCCA's impacts: Increasing awareness and creating data

A report on the effectiveness of the BCCA, published by the DGS in mid-2023, concluded that there was not yet enough information available to assess the effectiveness of the act in reducing emissions. However, by that time, 65 manufacturing facilities had produced EPDs, 54 of which were compliant with the relevant GWP limit.

But, interviewees stressed, the key impact to date has been about raising awareness and collecting data. "It's really been a foundational piece of policy in terms of increasing awareness," says Lewis at the CLF. "It has moved the building industry in the United States to deeply embed the notion of embodied carbon in its planning and development," agrees Williams at the Center for American Progress.

It has led to a substantial increase in the capacity of manufacturers to measure carbon emissions and the volume of that data from those manufacturers, adds Tilak at the RMI. "There's been a dramatic increase in transparency of embodied carbon impact," he says. Such transparency would be valuable for Indian policymakers in informing broader climate policy, and would be of value to Indian steelmakers in selling to clients keen to understand and reduce their Scope 3 (indirect) emissions.

Lewis at CLF notes that this process has been supported by the development of the EC3 Tool by a non-profit in collaboration with nearly 50 industry partners. It offers an EPD database and a means for the building industry to compare, specify and procure lower-carbon materials, including steel.

EPD data collection can be leveraged for other supply chain transparency initiatives, says Lewis. "If you have other goals beyond the environment, it gives you the opportunity to tie these together," she says. These could include employment practices or supply chain information.

There are now nearly 200 EPDs across the steel sector in the United States. The data contained within EPDs can also be leveraged for use by private sector buyers, such as developers and companies with commitments to reduce their Scope 3 emissions from building their facilities (like Microsoft and other tech companies). The LEED rating system also rewards demonstrating reductions using EPDs, and assessing embodied carbon is a requirement on all projects in the new rating system, Lewis adds.

The BCCA also began to get officials within government comfortable with how to include EPDs in their procurement processes, adds Tilak at the RMI. It has informed the creation of Buy Clean legislation in a further eight states in the United States, and the development of a Federal buy-clean programme by the Biden Administration. In a December 2021 Executive Order, it launched a Buy Clean task force to develop a Federal buy clean policy. It issued a set of federal-state buy clean principles in December 2023, but had not developed a federal policy before losing the 2024 election.



Lessons for India

As India develops its own green public procurement policy for low-carbon steel, the lessons and learnings from California's BCCA can inform that policy development. Since the introduction of the BCCA, the EPD has become further entrenched in green procurement practice around the world. It provides a standardised, widely applicable tool for informing green procurement as well as generating data and insights for other purposes.

Given that public procurement forms a large share of overall steel demand in India, it's a powerful tool to drive decarbonisation in the sector. The criteria set in India's green steel taxonomy can encourage the use of best available technology and have those costs covered by the premium that public procurement could provide. However, for large capex investments for significantly lower-carbon steel, a long-term path indicating the evolution of the green steel taxonomy will be vital for companies to undertake investment decisions in the coming years. Indian steel is in a capex cycle, which provides a window of opportunity for ambition in low-carbon steel investments, but which also requires longer-term clarity, if that opportunity is to be grasped.

If India aims to enhance the export competitiveness of its low-carbon steel, adopting EPDs for select facilities could help unlock access to broader international markets. Companies like Tata Steel have already secured the country's first EPDs, laying the groundwork for an ecosystem where export-oriented firms can leverage India's advantage in low-cost renewable electricity. This is particularly relevant for facilities using scrap and electric arc furnaces (EAFs), which offer a more cost-effective decarbonisation pathway than conventional steel-making technologies.

The way that BCCA was introduced, and how it operates, also holds important lessons for India. Bringing workers, companies and industry associations into the fold proved invaluable in reaching consensus around the legislation. The development of tools to help data collection is important. However, it is important to acknowledge the fundamentally different context of the BCCA in California and the Green Steel Taxonomy in India.

Most of the steel in California and the United States is made via scrap steel and electric arc furnaces, while most Indian steel is made via blast furnaces and induction furnaces, using limited scrap. While BCCA does not differentiate between primary and secondary steel, incorporating this differentiation will be vital for consistency and clarity in India.

The BCCA also shows that institutions play a key role in successful policy. India's NISST will play a vital role in collecting feedback and engaging industry as it embarks on certification and data collection. The information it collects will be vital for future iterations.

Interviewees

This case study is based on desk research and interviews with the following low-carbon procurement experts:

- Nancy Gillis, a Zurich-based industrial decarbonisation expert
- Meghan Lewis, program director at the Carbon Leadership Forum (CLF), in Seattle, Washington State
- Anish Tilak, at the RMI's Carbon-free Buildings Program
- Chandler Randol, a senior associate at the Rocky Mountain Institute (RMI) in Washington, DC
- Mike Williams, a senior fellow at the Center for American Progress

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